

Closed Native Verification Systems (CNVS): Complete Formal Theory and Emergent Security Theorem

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Abstract

This paper introduces Closed Native Verification Systems (CNVS), a formal class of verification systems in which state membership is verification-dependent rather than primitive. The theory combines a typed structural universe, a type-preserving recursive decomposition, a random assignment of non-uniform fragments, limited verifier knowledge, and a global invariant check, first applying data convergence in a local verification environment. A conditional probabilistic theorem limits unauthorized reconstruction with a binomial tail and produces exponential suppression when the reconstruction threshold exceeds the expected number of useful compromised fragments. An asymptotic scaling theorem shows that security can increase as the system expands in the presence of explicit assumptions. The technical difficulty of reconstruction outside the system arises from the strictly local knowledge available to each verifier.

1. Primitive Domains and Typed Universe

Let $B = \{0,1\}$ and B^ be the set of all finite binary strings. Define the primitive domains:*

$Id \subseteq B^$ (identifiers),*

$D \subseteq B^$ (declared data),*

$At \subseteq B^$ (attribute classes),*

$Val \subseteq B^$ (externally reported values),*

P (finite set of logical properties),

τ (finite set of structural types).

The structural universe is:

$U = Id \times D \times At \times Val \times Pow(P)$.

Each element $s \in U$ has the form $s = (id, d, a, v, p)$.

2. State Space

A system state at time t is:

$S(t) = (E(t), R(t), C, V_L)$,

where $E(t) \subseteq U$ is the set of admitted elements, $R(t) \subseteq E(t) \times E(t)$ is a directed relation set, $C = \{c_1, \dots, c_m\}$ is a finite family of invariant predicates, and V_L is the local verification function.

3. Local Verification

The local verification function is:

$$V_L : Id \times D \times At \times Val \rightarrow \{0,1\} \times Pow(P).$$

For $x = (id, d, a, v)$, let:

$$V_L(x) = (b, p),$$

where

$$b = Form(id,d,a,v) \wedge Conv_a(d,v).$$

Local admissibility is:

$$Adm_L(s,t) = 1 \Leftrightarrow \pi 1(V_L(s)) = 1.$$

4. Recursive Decomposition and Reconstruction

The decomposition operator is:

$$D_t : U \rightarrow Pow_fin(U).$$

For every admissible element s :

$$D_t(s) = \{F1, ..., Fk\}, 1 \leq k < \infty.$$

Type preservation:

$$\forall s' \in D_t(s), type(s') = type(s).$$

Reconstruction:

$$Rec_t(D_t(s)) = s$$

up to canonical encoding.

5. Non-Uniform Fragmentation

There exist fragments Fa and Fb such that:

$$|I(Fa)| \neq |I(Fb)|.$$

Hence decomposition is intentionally non-uniform. Some fragments contain more information than others.

6. Knowledge Restriction

For each external verifier v_j assigned to attribute a_j :

$$I_min(a_j) \leq K_V(j,t) \leq I_N(i,t) \leq I_G(t)$$

where $I_{\min}(a_j)$ is the minimum information required to preserve the semantic meaning of the assigned verification task.

Knowledge cannot converge to zero absolutely, since complete loss of information would eliminate the verifier's ability to perform the assigned task.

7. Information Density

For a fragment F_j :

$$\rho_N(j,i,t) = |I_F(j,t)| / |I_N(i,t)|,$$

$$\rho_G(j,t) = |I_F(j,t)| / |I_G(t)|,$$

$$\rho_C(j,i,t) = \rho_N(j,i,t)^\alpha \rho_G(j,t)^\beta,$$

where $\alpha, \beta \in [0,1]$ and $\alpha + \beta = 1$.

8. Random Assignment

Terminal fragments are assigned to external verifiers by:

$$A_t : T(t) \times \Xi \rightarrow V_{\text{ext}}(t),$$

where Ξ is a probability space and $T(t)$ is the set of terminal fragments.

9. Axiom I: Verification-Dependent Existence

For every element s and time t :

$$s \in E(t) \Rightarrow \text{Adm}_L(s,t) = 1.$$

Existence inside the system is a derived consequence of successful verification.

10. Axiom II: Randomized Non-Uniform Fragmentation

Every admissible element is recursively decomposed into non-uniform fragments that are assigned randomly:

$$D_t(s) = \{F1, \dots, Fk\},$$

$\exists Fa, Fb : |I(Fa)| \neq |I(Fb)|,$
 $A_t : T(t) \times \mathcal{E} \rightarrow V_{\text{ext}}(t),$
 $\text{Rec}_t(D_t(s)) = s.$

11. Axiom III: Non-Reducibility of Global Validity

Global validity is not reducible to local validity:

$V_G(S) \neq \bigwedge_{s \in E} V_L(s).$

Instead:

$V_G(S) = \text{LocalOK}(E) \wedge \text{Cons}_R(R, E) \wedge \text{Inv}_C(S).$

11a. State Transition Law

A candidate successor state is admitted only if it satisfies global verification:

$S(t+1)$ exists in the system if and only if $V_G(S(t+1)) = 1.$

Equivalently, any candidate state S' such that $V_G(S') = 0$ is rejected and cannot become a valid system state.

Thus, state evolution is verification-dependent and constrained by all structural invariants.

12. Fundamental Lemmas

Lemma 1. Local admissibility is necessary for state membership.

Lemma 2. Local validity is insufficient for global validity.

Lemma 3. Type is preserved across decomposition depth.

Lemma 4. Reconstruction conserves payload information.

Lemma 5. Metadata overhead permits encoded expansion.

Lemma 6. Average fragment density decreases under controlled subdivision.

13. Principle of Knowledge Restriction

For every external verifier:

$$I_{\min}(a_j) \leq K_V(j,t) \ll I_G(t).$$

If K_V remains bounded while $|I_G|$ grows, then:

$$|K_V(j,t)| / |I_G(t)| \rightarrow 0.$$

Thus each verifier retains the minimum semantic information required to perform the assigned task while controlling an asymptotically vanishing fraction of the total global information.

Equivalently, the verifier's knowledge provides negligible information about the native information content, so that

$$H(I_N | K_V(j,t)) \approx H(I_N),$$

or, equivalently,

$$I(I_N ; K_V(j,t)) \approx 0.$$

14. Probabilistic Security Model

Let:

k = number of terminal fragments,

r = reconstruction threshold,

C = colluding verifier count,

Q = total verifier count,

$$q = C/Q,$$

λ_j = probability that a compromised fragment is useful.

Define $X_j = 1$ if fragment j is compromised and useful, and

$$X = \sum_j X_j.$$

Unauthorized reconstruction occurs when:

$$Rec^* = \{X \geq r\}.$$

$$\omega(D_i) = I(G; F_i) / H(G)$$

$$\lambda_{\max} = \max_j \lambda_j$$

$$\lambda_j = \Pr(\omega(D_i) \geq \delta)$$

$$p_{\text{eff}} = q \lambda_{\max}$$

$$\tau \in (0, 1),$$

where τ is the minimum normalized amount of useful information required for unauthorized reconstruction.

15. Conditional Security Theorem

If X is stochastically dominated by $\text{Bin}(k, p_{\text{eff}})$, then:

$$\Pr(\text{Rec}^*) \leq \sum_{i=r}^k \binom{k}{i} p_{\text{eff}}^i (1-p_{\text{eff}})^{k-i}.$$

Proof.

By definition, unauthorized reconstruction requires at least r useful compromised fragments.

Under stochastic domination, $X \leq_{\text{st}} Y$ where $Y \sim \text{Bin}(k, p_{\text{eff}})$. Therefore:

$$\Pr(X \geq r) \leq \Pr(Y \geq r),$$

which equals the stated binomial upper tail. QED.

16. Exponential Chernoff Bound

Let $\mu = k p_{\text{eff}}$. If $r > \mu$, then:

$$\Pr(\text{Rec}^*) \leq \exp(-(r-\mu)^2 / (2\mu + (r-\mu))).$$

Proof.

Apply the multiplicative Chernoff bound to $Y \sim \text{Bin}(k, p_{\text{eff}})$. QED.

17. Emergent Security Scaling Theorem

Let n denote system scale. Assume:

$$k(n) \rightarrow \infty,$$

$p_{\text{eff}}(n)$ remains bounded or decreases,

$$r(n) - k(n)p_{\text{eff}}(n) \rightarrow +\infty.$$

Then:

$$\Pr(\text{Rec}^*_n) \rightarrow 0.$$

Proof.

Substitute the scaling assumptions into the Chernoff exponent. The exponent diverges to $+\infty$, hence the probability tends to zero. QED.

In the weighted formulation, if

$$q \bar{\omega} < \tau,$$

where q is the probability that a fragment is assigned to a colluding verifier and $\bar{\omega}$ is the average informational weight of compromised fragments, then:

$$q \bar{\omega} < \tau \implies \lim_{n \rightarrow \infty} P[\sum_{i=1}^n Y_i \omega(D_i) \geq \tau n] = 0.$$

17a. Generalized Emergent Security Scaling under Dependent Collusion

The Chernoff analysis presented above assumes independent compromise events and provides an idealized asymptotic approximation. A more general formulation considers coordinated colluding adversaries and conditional compromise probabilities.

Let:

- q denote the probability that a critical fragment is assigned to colluding verifiers;
- p_{inf} denote the probability that a critical fragment not directly controlled by colluding verifiers can nevertheless be inferred from compromised information;
- m denote the number of critical low-inference fragments required for unauthorized reconstruction.

Define the single-fragment compromise probability as

$$p_{\text{comp}} = q + (1 - q)p_{\text{inf}}.$$

Let C_i denote the event that the i -th critical fragment is compromised.

Assume that, for every critical fragment,

$$P(C_i | C_1, \dots, C_{i-1}) \leq p_{\text{comp}},$$

with

$$0 \leq p_{\text{comp}} < 1.$$

Let Rec^* denote the event of unauthorized full reconstruction.

Theorem 17a.1 (Generalized Emergent Security Scaling)

Under the above assumptions,

$$P(\text{Rec}^*) \leq p_{\text{comp}}^m.$$

Proof.

Unauthorized reconstruction requires compromise of all m critical fragments:

$$\text{Rec}^* = C_1 \cap C_2 \cap \dots \cap C_m.$$

By the chain rule of probability,

$$P(\text{Rec}^*) = \prod_{i=1}^m P(C_i | C_1, \dots, C_{i-1}).$$

Using the uniform conditional bound,

$P(\text{Rec}^*) \leq \prod_{i=1}^m p_{\text{comp}} = p_{\text{comp}}^m.$
QED.

Corollary 17a.1 (Asymptotic Security)

*If $0 \leq p_{\text{comp}} < 1$, then
 $P(\text{Rec}^*) \rightarrow 0$ as $m \rightarrow \infty$.*

Corollary 17a.2 (Non-Inferable Critical Fragments)

*If
 $p_{\text{inf}} = 0$,
then*

*$p_{\text{comp}} = q$
and therefore
 $P(\text{Rec}^*) \leq q^m.$*

Design Formula.

Given a target attack probability η , with $0 < \eta < 1$, the minimum number of critical low-inference fragments required to satisfy

$P(\text{Rec}^) \leq \eta$*

is

$m_{\text{min}} = \lceil \ln(\eta) / \ln(p_{\text{comp}}) \rceil,$

for $0 < p_{\text{comp}} < 1$

Since $\eta \in (0,1)$ and $0 < p_{\text{comp}} < 1$, both $\ln(\eta)$ and $\ln(p_{\text{comp}})$ are negative. Therefore the ratio $\ln(\eta) / \ln(p_{\text{comp}})$

is well defined and strictly positive. The condition $p_{\text{comp}}^m \leq \eta$

is equivalent to $m \ln(p_{\text{comp}}) \leq \ln(\eta)$. Since $\ln(p_{\text{comp}}) < 0$, division by $\ln(p_{\text{comp}})$ reverses the inequality, yielding $m \geq \ln(\eta) / \ln(p_{\text{comp}})$. Therefore,

$m_{\text{min}} = \text{ceil}(\ln(\eta) / \ln(p_{\text{comp}})).$

Remark 17a.1 (Semantic lower bound and practical scaling)

The asymptotic expression $m \rightarrow \infty$ should be interpreted as an idealized scaling limit.

In practical CNVS systems, fragmentation cannot proceed indefinitely, because terminal fragments must preserve enough semantic content for honest verification agents to perform their assigned tasks.

Therefore, the effective fragmentation depth is constrained by the semantic lower bound

$$I_F \geq I_{\min},$$

where $I_{\min}$ denotes the minimum information required for task comprehension.

The practical design problem is not to maximize fragmentation without limit, but to choose a configuration in which the number of critical low-inference fragments m is sufficiently large to satisfy

$$P(\text{Rec}^*) \leq \eta,$$

while every terminal fragment still satisfies

$$I_F \geq I_{\min}.$$

Thus, the security condition becomes a constrained design problem:

maximize epistemic restriction subject to semantic verifiability.

Equivalently, the system should increase m only up to the point where honest verifiers remain capable of local task execution, while adversarial coalitions remain unable to reconstruct the global state.

A practical CNVS configuration is feasible if

$$m_{\max} \geq m_{\min},$$

where $m_{\max}$ denotes the maximum number of fragments compatible with the semantic lower bound $I_F \geq I_{\min}$, and $m_{\min}$ denotes the minimum number of critical low-inference fragments required to satisfy the target security condition $P(\text{Rec}^) \leq \eta$.*

This feasibility condition states that the system can achieve the desired security level while preserving semantic verifiability.

Remark 17a.2 (Entropic interpretation of inference probability)

The inference probability p_{inf} can be interpreted as a function of the residual uncertainty of the global state. Let

$$K_{\text{adv}} = \sum_{i \in A} \omega(D_i)$$

denote the normalized informational weight controlled by the adversarial coalition. As K_{adv} increases, the residual uncertainty about the global state decreases, and the probability of inferring non-controlled fragments may increase. Therefore one may write

$$p_{\text{inf}} = v(K_{\text{adv}}),$$

where v is a non-decreasing function. The generalized compromise probability becomes

$$p_{comp} = q + (1-q)v(K_{adv}).$$

Thus,

$$P(Rec^*) \leq [q + (1-q)v(K_{adv})]^m.$$

This formulation connects the dependent-collusion model with the entropy-based interpretation of fragment utility. It does not replace the generalized security bound, but gives an informational interpretation of the inference term.

Interpretation.

This result generalizes the original Emergent Security Scaling Theorem by removing the independence assumption. It shows that exponential security scaling remains valid under coordinated collusion, provided that the conditional compromise probability of each critical fragment is uniformly bounded by a constant strictly smaller than one.

18. Interpretation

Under the stated assumptions, system growth increases fragmentation and epistemic restriction. As a result, unauthorized reconstruction becomes exponentially less probable.

Security is therefore an emergent property of scale.

In an idealized uniform model, the number of possible fragment assignments grows factorially with the number of terminal fragments. The corresponding combinatorial entropy is

$$H_n = \log_2(k(n)!).$$

Therefore, if $k(n) \rightarrow \infty$, then $H_n \rightarrow \infty$, indicating that the uncertainty faced by an external adversary increases as the system expands.

18a. Corollary: Global Veto Property

If at least one essential contribution to a global invariant remains unknown or uncontrolled by colluding verifiers, reconstruction of a globally valid state is not guaranteed.

Any incorrect estimation of that contribution may cause one or more global invariants to fail, yielding:

$$V_G(S) = 0$$

and rejection of the candidate state.

18b. Critical Fragment Principle

The Security of the System does not depend solely on the percentage of Verifiers under adversarial control, but on whether the adversary possesses all essential information required to reconstruct a globally valid state.

A Fragment is said to be essential if its absence prevents the construction of at least one globally admissible configuration satisfying the System invariants.

A Fragment is said to be non-inferable if it cannot be derived from the information already available to the adversary.

Formally, let K_{adv} denote the set of Fragments known to the adversary. If there exists at least one Fragment F_h such that

$F_h \notin K_{adv}$

and F_h is both essential and non-inferable, then successful reconstruction of a globally valid state is not guaranteed.

Equivalently,

$\exists F_h \notin K_{adv}$ such that F_h is essential and non-inferable $\implies \Pr(\text{Rec}^) < 1$.*

This principle emphasizes that the decisive factor is not the absolute proportion of colluding Verifiers, but the existence of at least one indispensable Fragment that remains outside adversarial control.

Consequently, even extensive collusion does not imply certain success unless the adversary acquires every essential Fragment required for globally coherent reconstruction.

18c. Gold Allocation Example

Suppose 100 orders should sum to 100 grams of gold.

If colluding agents know 99 orders totaling 99 grams, the remaining unknown order may represent any admissible quantity.

An incorrect estimate may produce a total inconsistent with the invariant, for example 102 grams instead of 100 grams.

Therefore:

$V_G(S) = 0$.

As the value of $|I_G|$ increases, the difficulty of reconstructing the missing information while preserving global invariant consistency increases correspondingly.

19. Scientific Positioning

CNVS combines concepts from information theory, probability theory, cryptography, formal methods, graph theory, and epistemic logic into a unified mathematical framework.

20. Open Problems

1. *Formal leakage models.*
2. *Dynamic collusion models.*
3. *Cryptographic reductions.*
4. *Information-theoretic lower bounds.*
5. *Empirical validation.*

Appendix A. Minimal Formal Definition

A structure

$M = (U, E, R, C, D, A, V_L, V_G, Rec, K)$

is a Closed Native Verification System if and only if:

1. *U is a typed finite-binary universe.*
2. *Membership requires local admissibility.*
3. *Global validity requires relation consistency and invariants.*
4. *Decomposition is finite, recursive, and reconstructible.*
5. *Verifier knowledge is strictly restricted.*
6. *Terminal assignments are randomized.*
7. *Invalid elements and states are rejected.*

END DOCUMENT

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